

Analyzing disparities in transit-based healthcare accessibility in the Chicago Metropolitan Area

Dong Liu 

Department of Geography and Geographic Information Science, University of Illinois at Urbana-Champaign
Human Environments Analysis Laboratory, Department of Geography and Environment, The University of Western Ontario

Mei-Po Kwan 

Department of Geography and Resource Management and Institute of Space and Earth Information Science, The Chinese University of Hong Kong
Department of Human Geography and Spatial Planning, Utrecht University

Zihan Kan 

Institute of Space and Earth Information Science, The Chinese University of Hong Kong

Key Messages

- White- and Hispanic-majority neighbourhoods with higher-than-average percentages of car-free households have the highest and lowest levels of healthcare accessibility respectively.
- The percentage of car-free households and minorities are positively and negatively related to the healthcare accessibility score respectively.
- The racial/ethnic factor has a stronger association with healthcare accessibility outcome than car ownership status.

Public transit is indispensable for car-free households to access healthcare. Meanwhile, different households have unequal transit-based healthcare accessibility due to different socio-economic factors such as race/ethnicity and car ownership. Few studies have comprehensively explored the inequality in transit-based healthcare accessibility by integrating both racial/ethnic inequity and car ownership. This study fills the gap with an up-to-date analysis of transit-based healthcare accessibility in the Chicago Metropolitan Area at the census tract level. The results show that the percentage of car-free households is positively related to transit-based healthcare accessibility; while a higher percentage of minorities (i.e., Black/Hispanic) is negatively related to transit-based healthcare accessibility. Among all neighbourhoods with higher percentages of higher-than-average car-free households, Hispanic-majority neighbourhoods fare the worst; while White-majority neighbourhoods have much better healthcare accessibility than both Black-majority and Hispanic-majority neighbourhoods.

Keywords: public transit, healthcare accessibility, car-free household, Chicago

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Correspondence to/Adresse de correspondance: Mei-Po Kwan, Department of Geography and Resource Management and Institute of Space and Earth Information Science, The Chinese University of Hong Kong, Shatin, Hong Kong.

Email/Courriel: mpk654@gmail.com

Analyse des disparités en matière d'accessibilité aux soins de santé par les transports en commun dans la région métropolitaine de Chicago

Les transports en commun sont indispensables pour permettre aux ménages sans voiture d'accéder aux soins de santé. Cependant, l'accessibilité aux services de santé par les transports en commun est inégale d'un ménage à l'autre en raison de différents facteurs socioéconomiques tels que la race ou l'ethnie et la possession d'une voiture. Peu d'études ont exploré de manière exhaustive l'inégalité d'accès aux soins de santé par le biais du transport en commun en intégrant à la fois l'inégalité raciale ou ethnique et la possession d'une voiture. Cette étude tente de combler cette lacune en présentant une analyse actualisée de l'accessibilité aux services de santé par le transport en commun dans la région métropolitaine de Chicago, à l'échelle des secteurs de recensement. Les résultats montrent que le pourcentage de ménages sans voiture est directement proportionnel à une bonne accessibilité aux soins de santé par le transport en commun, tandis qu'un pourcentage plus élevé de minorités (c'est-à-dire Afroaméricains ou Hispaniques) est inversement proportionnel à une forte accessibilité aux soins de santé par le transport en commun. Parmi tous les quartiers où le pourcentage de ménages sans voiture est supérieur à la moyenne, les quartiers à majorité hispanique sont les plus mal pourvus en matière d'accès aux services, tandis que les quartiers à majorité blanche ont un bien meilleur accès à ceux-ci que les quartiers à majorité afroaméricaine ou hispanique.

Mots clés : transport en commun, accessibilité aux soins de santé, ménage sans voiture, Chicago

Introduction

Public transit is an indispensable part of the transportation systems in many cities as a substantial number of urban households do not own a car due to unaffordability (Tomer 2011). For those who do not own any cars, public transit is their only transport mode for accessing key social services such as healthcare. Existing studies have found that analysis of the spatial distribution patterns of healthcare can also offer insights into the patterns of health outcomes (Graves 2008). In metropolitan areas with high residential and economic segregation, residents of the same race/ethnicity tend to live in the same neighbourhoods or cluster together (Florida and Mellander 2015). Existing studies have examined healthcare access inequity with respect to segregated minorities (Dai 2010; Richardson and Norris 2010; Caldwell et al. 2017), urban-rural neighbourhoods (Sibley and Weiner 2011; Zhang et al. 2017), and car ownership (Páez et al. 2010). However, few studies have examined transit-based healthcare access disparities by considering race/ethnicity and car ownership together, especially in the context of the United States given its racial segregation history and car-dominated culture.

Our study fills this research gap by exploring transit-based healthcare disparities by considering race/ethnicity and car ownership. We study the inequity in transit-based healthcare accessibility at the level of the census tract (CT) and address how

race/ethnicity and car ownership affect transit-based healthcare accessibility in the context of the United States.

Our paper is structured as follows. The next section reviews the relevant healthcare accessibility literature. We then describe the study area, data sources, and adopted methodology before presenting the results and findings of our research. The final section provides a discussion and presents the conclusion of the study.

Review of existing research

Measures of healthcare accessibility

Healthcare accessibility is commonly defined as the ease of reaching healthcare services and facilities by overcoming certain travel impedances such as travel distance, time, and cost (Hewko et al. 2002; Higgs 2004; Mao and Nekorchuk 2013). Measures of healthcare accessibility include distance-based measures (Brabyn and Skelly 2002; Apparicio et al. 2008), gravity-based measures (Schuurman et al. 2010; Xiong et al. 2016), cumulative opportunity measures (Paez et al. 2010), the kernel density method (Guagliardo et al. 2004), and various methods based on the floating catchment area (FCA) (Luo and Wang 2003; Langford et al. 2016; Shah et al. 2016; Pan et al. 2018; Chen and Jia 2019). Apparicio et al. (2008) used minimum travel time and distance to the nearest hospital based on the road network to estimate the

geographic accessibility of public hospitals in New Zealand. Schuurman et al. (2010) adopted a gravity-based measure that integrates the travel time impedance function to calculate the spatial access to primary healthcare in Nova Scotia, Canada. Paez et al. (2010) used a cumulative opportunity measure based on average trip length derived with a multivariate regression model to determine accessibility to healthcare facilities in Montreal for both senior and non-senior populations. Guagliardo et al. (2004) employed the Gaussian kernel density method to determine spatial accessibility to primary care providers for children in Washington, DC. Luo and Wang (2003) proposed a two-step FCA (2SFCA) method to measure spatial accessibility to healthcare in the Chicago metropolitan region.

Among all of the above-mentioned measures of healthcare accessibility, gravity-based measures and FCA-based methods are two approaches that have been commonly adopted in recent studies (Luo and Whippo 2012; Mao and Nekorchuk 2013). Gravity-based measures determine the potential of opportunities for interaction between origin locations (e.g., residential locations) and destination locations (e.g., locations of healthcare facilities) (Hansen 1959). Usually, gravity-based measures calculate healthcare accessibility through discounting healthcare opportunities (i.e., number of healthcare providers/facilities) by a distance-decay function that integrates travel time (Schuurman et al. 2010; Jamtsho and Corner 2014; Xiong et al. 2016). However, the parameter within the distance-decay function, which determines the discounting rate, can be arbitrary in the absence of prior empirical studies specific to the study area. Besides, traditional gravity-based measures often overlook the demand for healthcare resources. FCA-based methods are a special form of gravity-based measures and have been commonly implemented in the two-step form, known as the 2SFCA method (Luo and Wang 2003). Earlier implementation of the 2SFCA method evaluated healthcare accessibility by the ratio of healthcare supply and demand within catchment areas of fixed size (Luo 2004). Essentially, while traditional gravity-based methods take a positive approach, the 2SFCA method takes a normative approach (Páez et al. 2012). Later, the 2SFCA method was improved to dynamically determine the supply and demand catchment sizes, which can help reveal more details in the spatial variation of accessibility

compared to results based on catchments of fixed sizes (Luo and Whippo 2012; Chen and Jia 2019). The 2SFCA method now takes into consideration that patients may cross administrative boundaries for healthcare services and allows for non-adherence to administrative boundaries. Nevertheless, traditional 2SFCA methods have some limitations. First, as a dichotomous measure, this approach assumes that all locations outside the catchment have no access. Second, it assumes that all locations inside the catchment have equal access; it uses fixed catchment sizes and does not account for variability in terms of travel time tolerance to access healthcare for people living in different parts of an area—for example, central city versus suburb (Luo and Whippo 2012). Some studies have attempted to overcome some of these limitations of the traditional 2SFCA method. For example, a model-based approach, which allows for the adjustment of catchment size based on the characteristics of specific locations, was proposed by Paez et al. (2010) and implemented in Páez et al. (2010). In addition, intra-catchment spatial variability can be addressed by adopting non-binary distance-decay functions (Paez et al. 2019).

Inequity of healthcare accessibility

Disparities in health outcomes often occur among residents with different levels of access to healthcare, and spatial patterns of illness have been known to be associated with the distribution of healthcare (Graves 2008).

Spatial disparities in healthcare accessibility have been found at various spatial scales, including the city level (Luo and Wang 2003), the county level (Jin et al. 2015), the regional level (Luo and Qi 2009), the state/provincial level (Mao and Nekorchuk 2013; Evans et al. 2017; Ding et al. 2018; Huang et al. 2020), and the national level (Rosero-Bixby 2004; Chen et al. 2014; Zhang et al. 2018). However, inequity in healthcare accessibility not only has a spatial dimension, it can also have economic and racial/ethnic dimensions. For instance, Mayaud et al. (2019) examined the healthcare accessibility in Cascadia, which includes Seattle, Portland, and Vancouver, and found that a higher percentage of residents living in low-income neighbourhoods have difficulties accessing healthcare compared to those living in high-income neighbourhoods in those three cities. Chen et al. (2016) found disparities in healthcare

accessibility across the non-Hispanic White, non-Hispanic Black, and Hispanic populations in the United States. Copeland (2005) found major healthcare access inequity between White and Black Americans. These recent studies (Copeland 2005; Chen et al. 2016; Mayaud et al. 2019) confirmed the substantial associations between economic status or race/ethnicity and healthcare access inequity.

Further, the inequity of healthcare accessibility has also been revealed between people using different transport modes. Due to decades of car-oriented transportation policies in the United States, the development of public transit has been lagging substantially behind the development of road networks, which exerts inequitable impacts on the carless minority and low-income populations and limits their access to various socioeconomic opportunities including healthcare service (Sanchez et al. 2004). Existing literature tends to examine inequity in healthcare accessibility among people using the same transport mode. For example, Bagheri et al. (2005) examined spatial accessibility to primary healthcare services for drivers in the Otago region, New Zealand, using the least-cost path analysis model. Mao and Nekorchuk (2013) studied spatial accessibility to healthcare facilities for drivers and bus riders in Florida by employing the 2SFCA method. In the context of Ghana, spatial accessibility to healthcare facilities was measured using a location-allocation model based on walking distance, given the lack of any transport mode for ordinary citizens in the country (Oppong and Hodgson 1994). However, past studies have not particularly examined transit-based healthcare accessibility by focusing on car-free households which are transit-dependent. Even in the car-dominated culture in the United States, many major urban areas still have a large number of car-free households that rely on public transit in the region (Tomer 2011; Liu et al. 2020; Liu et al. 2021b). Meanwhile, little is known about the transit-based healthcare accessibility for those car-free households in the context of the United States. Furthermore, due to the discrimination and segregation history in many major cities in the United States such as Chicago, non-White residents, who tend to be of low income and are less likely to own a car (Brown 2017), have largely been confined to reside in the central city where the public transit density tends to be higher (Liu and Kwan 2020). The spatial confinement of

non-White minorities to the central city and the sprawl of job opportunities to the suburban regions where transit density is low have led to lower job accessibility for the central city minorities, which has been called the spatial mismatch between jobs and housing (Kain 1968). However, when it comes to accessing healthcare resources, we know little whether carless non-White residents are disadvantaged compared to their White counterparts.

Therefore, in this study, we aim at exploring transit-based healthcare accessibility in the context of a major urban area in the United States by taking both race/ethnicity and car ownership into consideration to uncover how race/ethnicity and car ownership affect transit-based healthcare accessibility.

Data and methodology

Study area

The study area for this research is the Chicago Metropolitan Area (hereafter CMA), which includes seven counties (i.e., Cook, DuPage, Lake, McHenry, Kane, Kendall, Will) and is located in the northeastern region of Illinois (Figure 1a). The City of Chicago (hereafter Chicago) is the county seat of Cook County and also the most densely populated city in the study area (Figure 1b). The study area is the third-largest metropolitan area in the United States and has the second-largest public transit system in the country, which makes it a suitable study area in terms of analyzing the spatial inequality in transit-based healthcare accessibility. As of 2018, 6.37% of all households in the metropolitan area did not own a car (United States Census Bureau 2018). Additionally, the study area is known to have a high level of residential and socioeconomic concentration (Florida and Mellander 2015).

There are three transit providers in the study area, which include the Chicago Transit Authority (CTA), the Metra, and the Pace. The CTA is the largest transit operator in the study area providing both bus and rail services to neighbourhoods within Chicago and Cook County; the Pace operates suburban bus services connecting different suburban counties with each other as well as Chicago; and the Metra provides commuter rail service connecting suburban county residents with

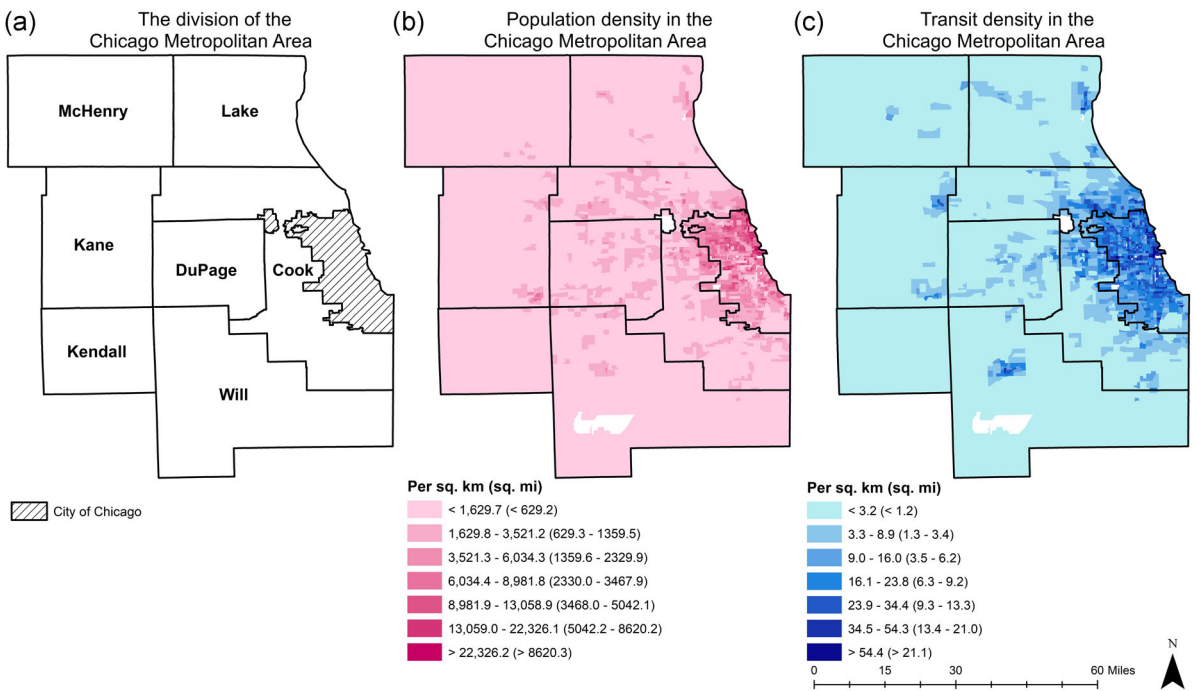


Figure 1
The (a) division, (b) population density, and (c) transit density in the CMA.

Chicago. As shown in Figure 1c, neighbourhoods within or near Chicago have a much higher transit density than the rest of the study area.

The distribution of car-free households, median household income, and majority race/ethnicity in the study area is shown in Figure 2a, b, and c, respectively, based on the Jenks natural breaks classification, which groups together similar values and maximizes the differences between classes.

Figure 2 shows that the CMA has a high degree of economic and residential concentration, which has been observed in existing literature (Florida and Meunier 2015). Specifically, most low-income neighbourhoods overlap the majority-Black/Hispanic neighbourhoods, which are mostly located in the southern and western parts of Chicago. The Black and Hispanic populations also account for the vast majority of the minority population and are commonly deemed as the most socially disadvantaged groups in the study area. In the meantime, the distribution of healthcare resources, which can be represented by healthcare workers (Wong et al. 2016; WHO World Health Organization 2019), presents the following pattern (Figure 3).

Figure 3 shows the distribution of healthcare workers per 1,000 people. It indicates that most healthcare workers are dispersed around the suburban counties, with a small concentration of healthcare workers in the centre of Chicago. This pattern reflects the fact that despite a few major hospitals (e.g., Rush University Medical Center, University of Illinois Medical Center, John H. Stroger Jr. Hospital of Cook County) being located within the Illinois Medical District near downtown Chicago, the overall distribution of healthcare workers in the study area signals that car-free households in Chicago are disadvantaged in that most healthcare resources are located outside Chicago where transit service tends to be sparse.

Data sources

The CT-level data used for our study mainly came from the American Community Survey (ACS), Census Transportation Planning Products (CTPP), and the Google Maps Directions API. The CT-level data include race/ethnicity, car ownership, healthcare worker, and travel time information. First, demographic (e.g., race/

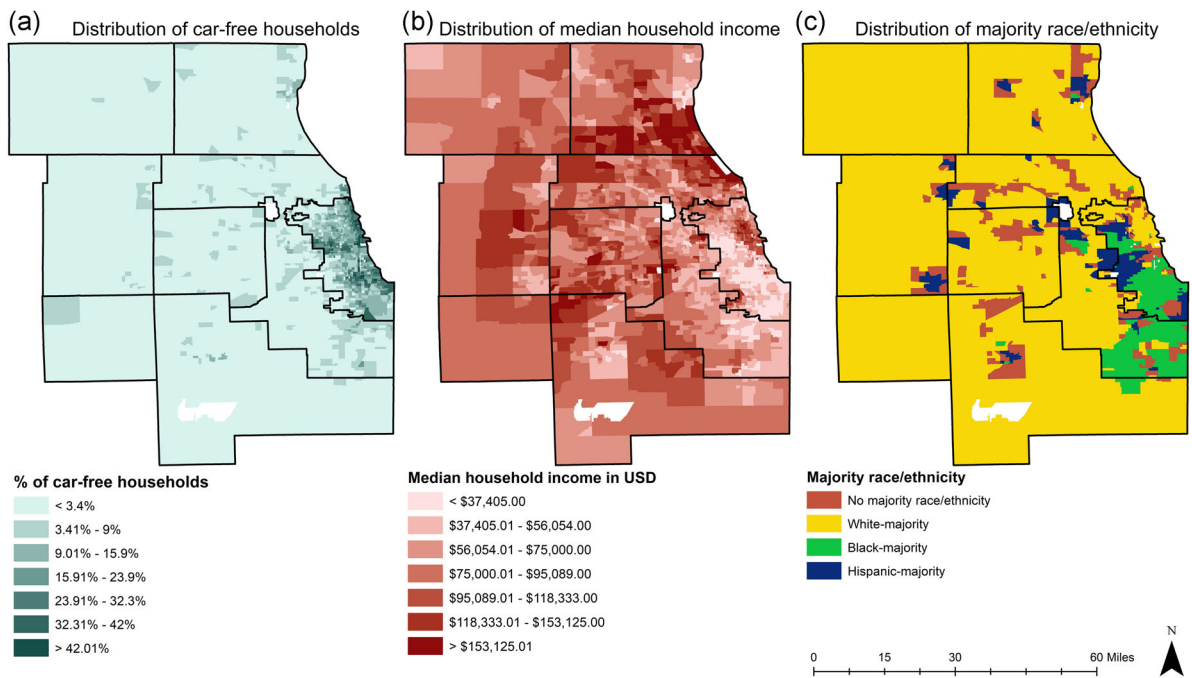


Figure 2
The distribution of (a) car-free households, (b) median household income, and (c) majority race/ethnicity in the CMA.

ethnicity) and car ownership information was derived from the 2018 Census data. When counting White and Black residents, we only counted non-Hispanic White and Black residents to avoid double-counting. Second, CT-level workplace-based healthcare worker information was derived from the latest CTPP data (i.e., the 2012–2016 5-Year CTPP data). Healthcare workers can be used to represent the distribution of healthcare resources (Wong et al. 2016; WHO World Health Organization 2019). Except for a few very recent studies (e.g., Paez et al. 2019), many traditional studies that considered healthcare facilities as healthcare resources did not distinguish healthcare facilities by their healthcare supply capacities (Loo and Lam 2012; Pan et al. 2018). Further, detailed data regarding the location of specific healthcare facilities, as well as their healthcare supply capacities (e.g., the number of physicians), are often not freely available in cities such as Chicago and need to be purchased for use (Luo and Wang 2003). By employing the number of healthcare workers in each CT, the healthcare supply capacity of each CT can be directly quantified and the healthcare supply-demand ratio can be calculated by dividing the number of healthcare

workers in a CT by the number of residents who can reach this CT after overcoming certain travel impedance (e.g., travel time). Transit-based travel time between CTs was gathered from the Google Maps Directions API, which calculates travel time based on real-time ingress/egress, transfer, waiting, and on-board time in the morning of a weekday and accounts for traffic conditions for the entire trip. Although previous studies (e.g., Widener et al. 2015) estimated travel time by also integrating ingress and egress walking, waiting, and transfer times, they did not account for traffic conditions for the entire trip. In comparison, the travel time obtained from the Google Maps Directions API includes real-time ingress/egress, transfer, waiting, and on-board time that account for traffic conditions for the entire trip. The travel time obtainment through Google Maps Directions API was implemented based on the documentation provided by Google Developers (2021).

Methodology

We adopt the gravity-based measure based on Shen (1998) to calculate transit-based healthcare

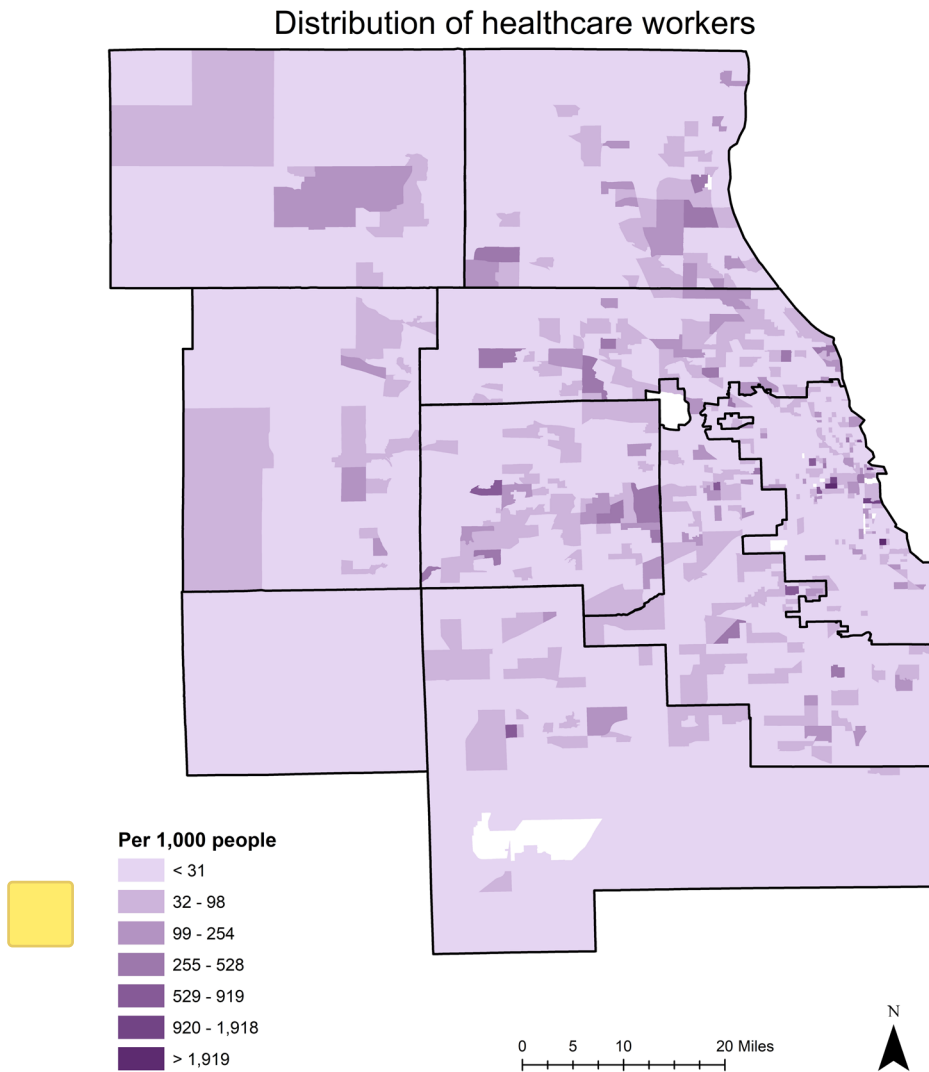


Figure 3
The distribution of healthcare workers in the CMA.

accessibility by integrating healthcare supply and healthcare demand, both of which are discounted by travel impedance. In order to account for the issue of the inflation effect as discussed previously by Delamater (2013), Luo (2014), and Wan et al. (2012), we modify the method based on Paez et al. (2019). As a result, healthcare demand is first calculated by Equations (1) and (2),

$$D_j = \sum_{r \in \{T_{rj} \leq T_0\}} \frac{P_r f(T_{rj})}{\sum_{r \in \{T_{rj} \leq T_0\}} f(T_{rj})} \quad (1)$$

$$f(T_{rj}) = \text{Exp}(-\beta T_{rj}) \quad (2)$$

where D_j is the transit-based potential demand for healthcare at CT j ; P_r is the population of

CT r whose centroid falls within the catchment (i.e., $T_{rj} \leq T_0$); T_{rj} is the transit-based travel time between CT r and CT j ; T_0 is the threshold transit-based travel time and its value is set at 50 minutes; $f(T_{rj})$ is the transit impedance function for measuring the spatial separation between CT r and CT j ; T_{rj} is the transit-based travel time between CT r and CT j ; and β is the impedance function parameter and its value is set at 0.15. The threshold travel time T_0 is determined based on Luo and Wang (2003), which measured healthcare accessibility for the same study area, in order to achieve spatial smoothing and reduce the variability of accessibility across space. The value for the impedance function parameter β is chosen because it is a relatively moderate rate and has a decay curve that declines neither too fast nor too slow. This choice is based on Kwan (1998), which compared and discussed different impedance function parameters.

After deriving the transit-based healthcare demand for each CT, we calculate the transit-based spatial accessibility to healthcare that integrates the transit-based healthcare demand based on Equations (3) and (4),

$$A_i = \sum_{j \in \{T_{ij} \leq T_0\}} \frac{H_j f(T_{ij})}{\sum_{j \in \{T_{ij} \leq T_0\}} f(T_{ij})} \quad (3)$$

$$f(T_{ij}) = \text{Exp}(-\beta T_{ij}) \quad (4)$$

where A_i is the transit-based accessibility for healthcare at CT i ; H_j is the number of healthcare workers of CT j whose centroid falls within the catchment (i.e., $T_{ij} \leq T_0$); D_j is healthcare demand of CT j whose centroid falls within the catchment (i.e., $T_{ij} \leq T_0$); T_{ij} is the transit-based travel time between CT i and CT j ; T_0 is the threshold transit-based travel time and its value is again set at 50 minutes; $f(T_{ij})$ is the transit impedance function for measuring the spatial separation between CT i and CT j ; T_{ij} is the transit travel time between CT i and CT j ; and β is the impedance function parameter and its value is again set at 0.15.

Equation (3) indicates that transit-based healthcare accessibility is measured by considering the number of healthcare workers weighted by healthcare demand and travel impedances. This two-step gravity measure takes into account both the

population demand for healthcare and travel impedance to healthcare when calculating spatial accessibility to healthcare, which reflects the barriers to healthcare access in terms of both distance and demand competition while accounting for the inflation effect.

In order to further test and examine the association between healthcare accessibility and car ownership and race/ethnicity, we also conducted multiple linear regression analysis using R. The dependent variable is the healthcare accessibility score of each CT, while the independent variables include the percentages of car-free households, Black population, and Hispanic population in each CT.

Results

The transit-based healthcare accessibility in the CMA was calculated based on Equations (1)–(4). After obtaining the healthcare accessibility results, we visualized the healthcare accessibility distribution as shown in Figure 4. The accessibility score ranges from 0.00 to 0.55. The highest scores are found mostly in CTs in DuPage and Cook Counties, especially in downtown Chicago; the lowest scores are mostly found in CTs on the periphery of Will and McHenry Counties. A high accessibility score indicates high transit-based healthcare accessibility and a low accessibility score indicates low transit-based healthcare accessibility.

As shown in Figure 4, within Chicago, most neighbourhoods have low healthcare accessibility except for some neighbourhoods in the northern part of the city. Most neighbourhoods with relatively high healthcare accessibility are located outside Chicago in the surrounding counties, especially in Cook and DuPage. Additionally, healthcare accessibility is distributed rather evenly within Cook and DuPage. In comparison, neighbourhoods in the southeast of Lake and McHenry, east of Kane, and northwest of Will have better healthcare accessibility compared to other neighbourhoods inside those respective counties. Neighbourhoods within Kendall in general have the worst healthcare accessibility in the entire study area. Overall, despite the higher transit density inside Chicago, as illustrated in Figure 1c, neighbourhoods in the south and west of Chicago are still disadvantaged in terms of

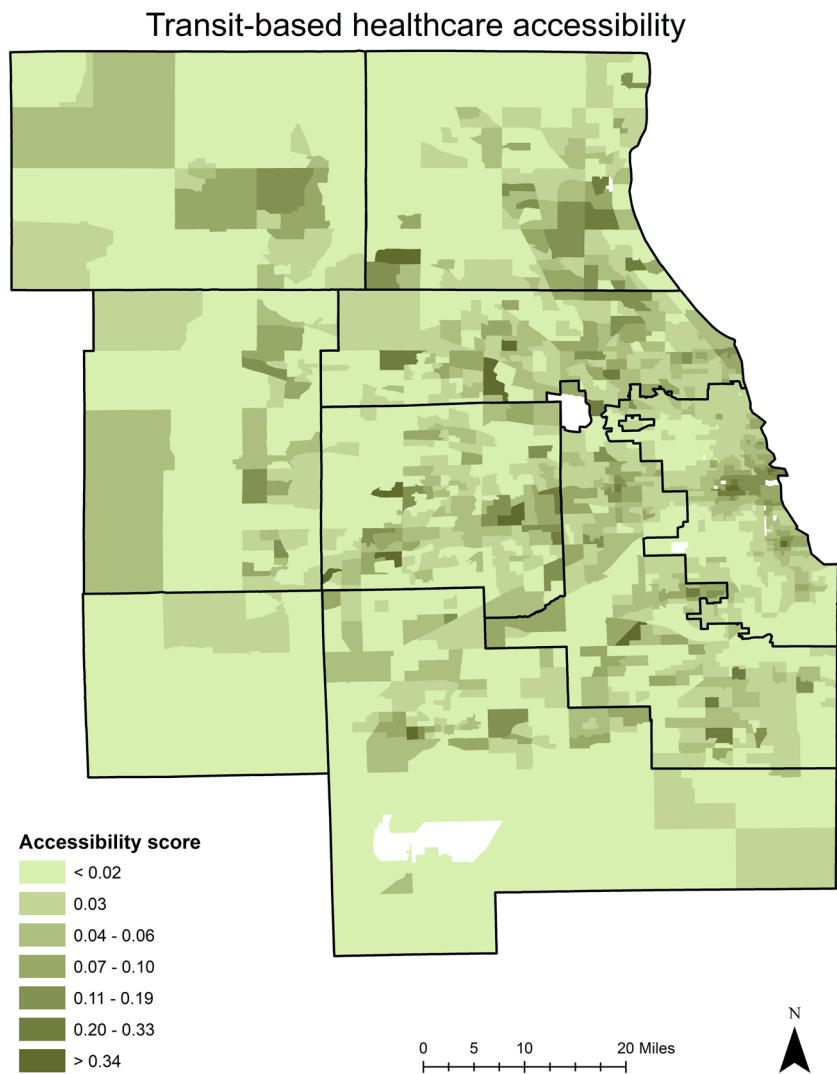


Figure 4
The distribution of transit-based healthcare accessibility in the CMA.

transit-based healthcare accessibility compared to many neighbourhoods outside Chicago. This may be explained by the distribution of healthcare workers outside Chicago, as shown in Figure 3, and the spatial proximity to healthcare workers for neighbourhoods outside Chicago. Meanwhile, there is also less competition for the healthcare workers for neighbourhoods outside Chicago due to lower population density outside Chicago, as shown in Figure 1b.

In order to illustrate the disparities in healthcare accessibility distribution by race/ethnicity and car ownership, we compared the results in Figure 4 and Figure 2. The comparison shows that majority-Black/Hispanic neighbourhoods in the west and south of Chicago generally have lower healthcare accessibility compared to the majority-White neighbourhoods to the north of Chicago, which indicates an evident race/ethnicity-based inequity in terms of healthcare accessibility. Nevertheless, it

can be seen that neighbourhoods dominated by minorities outside Chicago, especially in Lake, Kane, and Will, have better healthcare accessibility than their counterparts within Chicago. This may be explained by the fact that minority-dominated neighbourhoods in Lake, Kane, and Will are, on the one hand, spatially close to healthcare outside Chicago and, on the other hand, clustered around major Metra transit hubs including Waukegan Station, Elgin Station, Aurora Station, and Joliet Station, which provide them with good transit-based access. In comparison, although minority-dominated neighbourhoods also have high transit density, they are spatially further from the healthcare outside Chicago and face more competition in accessing the limited healthcare resources within Chicago due to higher population density as shown in Figure 1b.

As for the role of the percentage of car-free households, many of the neighbourhoods with high percentages of car-free households in the north of Chicago, which are mostly populated by wealthier White households, actually have good healthcare accessibility as shown by Figure 4 and Figure 2a. In comparison, neighbourhoods with high percentages of car-free households in the west and south of Chicago, which are mostly populated by poorer Black and Hispanic households, generally suffer from low healthcare accessibility. Outside Chicago, it can be seen that neighbourhoods with high percentages of car-free households mostly have relatively good healthcare accessibility, regardless of the dominant race/ethnicity.

To compare the overall healthcare accessibility score by race/ethnicity and car ownership, we divided all CTs in the study area into different racial/ethnicity (i.e., White/Black/Hispanic-majority) and car ownership (i.e., percentage of car ownership above/below average) categories.

As shown in Table 1, the overall accessibility score for all the CTs is 0.032. The score is 0.030 for all the CTs with higher-than-average car ownership percentages, and 0.037 for all the CTs with lower-than-average car ownership percentages. When only considering CTs with a White majority, the accessibility score for all the White-majority CTs is 0.035. It is 0.033 for White-majority CTs with higher-than-average car ownership percentages, and 0.049 for White-majority CTs with lower-than-average car ownership percentages. When only considering

Table 1
Comparison of healthcare accessibility score by race/ethnicity and car ownership.

	Total CTs	CTs with White Majority	CTs with Black Majority	CTs with Hispanic Majority
Total CTs	0.032	0.035	0.029	0.018
CTs with Car Ownership % Above Average	0.030	0.033	0.028	0.017
CTs with Car Ownership % Below Average	0.037	0.049	0.029	0.021

CTs with Black-majority, there are no substantial differences in the accessibility score regardless of the car ownership percentage. When only considering CTs with Hispanic-majority, the accessibility score for all the Hispanic-majority CTs is 0.037. The score is 0.031 for Hispanic-majority CTs with higher-than-average car ownership percentages, and 0.048 for Hispanic-majority CTs with lower-than-average car ownership percentages. Therefore, among CTs with lower-than-average car ownership percentages, White-majority neighbourhoods have much higher accessibility scores than Black-majority and Hispanic-majority neighbourhoods, while Hispanic neighbourhoods have the lowest accessibility scores among the three majority races/ethnicities. The same pattern is repeated among neighbourhoods with higher-than-average car ownership percentages—with White-majority neighbourhoods having the highest and Hispanic-majority neighbourhoods having the lowest transit-based healthcare accessibility scores respectively. Overall, neighbourhoods with lower-than-average car ownership percentages tend to have higher transit-based healthcare accessibility scores than neighbourhoods with higher-than-average car ownership percentages. Meanwhile, Black-majority and Hispanic-majority neighbourhoods, which are concentrated in the west and south of Chicago, tend to have substantially lower transit-based healthcare accessibility scores than the White-majority neighbourhoods. This pattern becomes more evident when compared to White-majority neighbourhoods with lower-than-average car ownership percentages that are concentrated in the north of Chicago as shown in Figure 2a.

Table 2
Regression results for healthcare accessibility and car ownership and race/ethnicity.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.037	0.002	21.290	0.000***
Car-free household %	0.001	0.000	7.557	0.000***
Black %	-0.021	0.004	-5.688	0.000***
Hispanic %	-0.033	0.004	-7.535	0.000***

*P ≤ 0.05. **P ≤ 0.01. ***P ≤ 0.001.

Moreover, the inequity in transit-based healthcare accessibility between the White-majority neighbourhoods in the north and the minority-dominated west and south also indicates that there are disparities in terms of the transit-based connection between Chicago and its surrounding counties where healthcare workers concentrate. Although Chicago as a whole has high transit density, the high transit density spans from the north of Chicago to the bordering neighbourhoods to the north and northwest of Chicago where there is a high concentration of healthcare workers, as shown by Figure 3. In comparison, the good transit density in the west and south does not span entirely to the bordering neighbourhoods to the south and southwest of Chicago. Moreover, the bordering neighbourhoods to the south and southwest of Chicago also appear to have lower concentrations of healthcare workers compared to the bordering neighbourhoods to the north and northwest of Chicago.

Overall, the result shows a complex distribution pattern of healthcare accessibility when considering the factors of car ownership and race/ethnicity. Next, we conducted further analysis based on the abovementioned multiple linear regression analysis, the results of which are shown in Table 2.

From Table 2, it can be seen that the percentage of car-free households is positively associated with healthcare accessibility, which means that a higher percentage of car-free households in a CT is associated with higher healthcare accessibility. The percentages of Black and Hispanic populations are negatively associated with healthcare accessibility, which means that a higher Black/Hispanic percentage in a CT is associated with lower healthcare accessibility. In addition, the racial/ethnic factor appears to have a stronger

association with healthcare accessibility than with car ownership status. The percentage of the Hispanic population in a CT has a stronger negative association with healthcare accessibility than the percentage of the Black population, which means that CTs with higher Hispanic percentages tend to fare worse in terms of transit-based healthcare accessibility. The p-values of all three independent variables show that they are significant.

In general, the results indicate that the transit service in the study area functions well in terms of providing transit-based access to healthcare to car-free White households, which are mostly located in the north of Chicago, but not so well for car-free minority (i.e., Black/Hispanic) households, which are mostly located in the west and south of Chicago. Minority-dominated neighbourhoods with low percentages of car-owning households suffer from much lower transit-based healthcare accessibility, with Hispanic-majority neighbourhoods having the worst transit-based healthcare accessibility outcome.

Discussion and conclusions

By taking care ownership and race/ethnicity into consideration, our study found transit-based healthcare access inequity in the CMA, which has a high degree of economic and residential concentration with most low-income neighbourhoods being majority-Black/Hispanic neighbourhoods in the west and south of Chicago. These poorer minority-dominated neighbourhoods have lower healthcare accessibility compared to the wealthier White-majority neighbourhoods in the north of Chicago. While the percentage of car-free households is positively associated with healthcare accessibility (i.e., a higher percentage of car-free households associate with a higher healthcare accessibility score), the racial/ethnic factor is negatively associated with healthcare accessibility (i.e., a higher percentage of the minority population associate with a lower healthcare accessibility score). The racial/ethnic factor also appears to have a stronger association with healthcare accessibility than with car ownership status. In addition, most neighbourhoods with high healthcare accessibility are actually located outside Chicago, which has a low transit density but also a low population density and spatial proximity to healthcare

resources. This signals that spatial proximity to and low competition for healthcare resources could provide an advantage in accessing healthcare to some degree.

Nevertheless, our findings do not discount the importance of transit service, especially given the good transit-based healthcare accessibility for the car-free households in the White-majority north of Chicago. In fact, policymakers should take note of the negative impact of the highway systems on the car-free households in the minority-dominated west and south parts of Chicago. The presence of various highways (e.g., Interstate 55, Interstate 90, and Interstate 290) cut right across the Black-majority and Hispanic-majority neighbourhoods in the south and west of Chicago, while largely staying clear of the White-majority neighbourhoods in the north of the city. This leads to the problem that although the west, south, and north of Chicago appear to have similar transit densities, residents living there have substantially different experiences when walking to the transit stations due to the presence/absence of interstate highways. In other words, it is much easier and less time-consuming for people from car-free households in the White-majority north to walk to the train stations (e.g., Armitage Station and Fullerton Station in the Lincoln Park neighbourhood; Diversey Station, Wellington Station, and Belmont Station in the Lake View neighbourhood; Argyle Station, Lawrence Station, and Wilson Station in the Uptown neighbourhood), which are located mostly along nice walkable business strips with light vehicular traffic. In contrast, people from car-free households in the minority-dominated west and south usually take more time to walk to the train stations (e.g., Cicero Station in the Austin neighbourhood; Pulaski Station and Kedzie-Homan Station in the Garfield Park neighbourhood; 47th Station and Garfield Station in the Fuller Park neighbourhood; 63rd Station in the Englewood neighbourhood; 79th Station in the Chatham neighbourhood), which are usually located along highway corridors such as Interstate 90 and Interstate 290 and can only be accessed via the noisy and dangerous overpasses or underpasses with heavy traffic. The CTA red line runs from the north to the south of Chicago, and reveals the contrast in ease of access to transit stations between the two parts of Chicago. The surrounding streets of Fullerton Station and Belmont Station in the wealthy White-majority Lincoln Park and Lake View

neighbourhoods in the north are much more walkable compared to the 47th Station and 63rd Station in the poor Black-majority Fuller Park and Englewood neighbourhoods in the south. Outside Chicago, the concentration pattern shows a vast majority of neighbourhoods being dominated by the Whites except the neighbourhoods within Cook to the northwest and south of Chicago as well as a few small concentrations in Lake, Kane, and Will. Nevertheless, outside Chicago, minority-dominated neighbourhoods with high percentages of car-free households generally perform well in terms of healthcare accessibility. This may be a function of the small concentrations of minority neighbourhoods with high car-free percentages outside Chicago being mostly clustered around major suburban transit hubs (e.g., Waukegan Station, Elgin Station, Joliet Station), which are located in pedestrian-friendly places with light vehicular traffic.

In addition to the pedestrian-hostile environment when accessing the transit stations in the west and south, we also found that the transit connections between the west and south to their immediate bordering neighbourhoods are worse compared to those in the north. This disadvantage the car-free households living in the west and south as they would have more difficulty accessing bordering neighbourhoods where there are higher concentrations of healthcare workers. Moreover, the bordering neighbourhoods for the west and south have lower concentrations of healthcare workers compared to the bordering neighbourhoods of the north, which further complicates the access to healthcare for the car-free households in the poor minority-dominated west and south.

In light of these findings, policymakers should make effort to, first of all, improve pedestrian accessibility to the transit stations in the west and south of Chicago. They should also open more transit connections between the west and south with their immediate bordering neighbourhoods which have higher healthcare worker concentrations. Moreover, more healthcare resources should be channeled directly towards the west and south of Chicago since the healthcare workers within Chicago are mostly concentrated in the north.

The long segregation history of the study area (Moore 2016) has been found responsible for a wide range of socio-economic injustice issues including opportunities for employment (e.g.,

Smith et al. 2021), education (e.g., Jankov and Caref 2017), and recreation (e.g., Liu et al. 2021a). The results of our study add another dimension to the injustice literature, in terms of healthcare access for the study area.

Our study analyzed the spatial distribution patterns of CT-level transit-based healthcare access in the CMA using the latest available census data in combination with transit travel time data. This empirical study provides policymakers with insights into spatial inequality in transit-based healthcare access that accounts for race/ethnicity and car ownership in a cost-effective and efficient manner and can be replicated in many other metropolitan areas in the country. However, due to the usage of the CT-level aggregate data, our study does not account for the variability of accessibility within each CT and is not immune from the modifiable areal unit problem. Future studies should seek to explore the intra-CT disparities of accessibility in order to provide a more fine-grained inequity analysis in healthcare accessibility.

Acknowledgments

Dong Liu was supported by a Marion G. Russell Graduate Fellowship from the University of Illinois at Urbana-Champaign. Mei-Po Kwan was supported by grants from the Hong Kong Research Grants Council (General Research Fund Grant no. 14605920; Collaborative Research Fund Grant no. C4023-20GF) and a grant from the Research Committee on Research Sustainability of Major Research Grants Council Funding Schemes of the Chinese University of Hong Kong. Zihan Kan was supported by an RGC Postdoctoral Fellowship from the Research Grants Council of Hong Kong (PDFS2021-4S08). The authors would like to thank the anonymous reviewers for their insightful comments, which helped improve the paper considerably.

References

- Apparicio, P., M. Abdelmajid, M. Riva, and R. Shearmur. 2008. Comparing alternative approaches to measuring the geographical accessibility of urban health services: Distance types and aggregation-error issues. *International Journal of Health Geographics* 7(1): 7.
- Bagheri, N., G. L. Benwell, and A. Holt. 2005. Measuring spatial accessibility to primary health care. Paper presented at the annual colloquium of the Spatial Information Research Centre, November 24th–25th, University of Otago, Dunedin, New Zealand.
- Brabyn, L., and C. Skelly. 2002. Modeling population access to New Zealand public hospitals. *International Journal of Health Geographics* 1(1): 3.
- Brown, A. E. 2017. Car-less or car-free? Socioeconomic and mobility differences among zero-car households. *Transport Policy* 60: 152–159.
- Caldwell, J. T., C. L. Ford, S. P. Wallace, M. C. Wang, and L. M. Takahashi. 2017. Racial and ethnic residential segregation and access to health care in rural areas. *Health & Place* 43: 104–112.
- Chen, X., and P. Jia. 2019. A comparative analysis of accessibility measures by the two-step floating catchment area (2SFCA) method. *International Journal of Geographical Information Science* 33(9): 1739–1758.
- Chen, J., A. Vargas-Bustamante, K. Mortensen, and A. N. Ortega. 2016. Racial and ethnic disparities in health care access and utilization under the Affordable Care Act. *Medical Care* 54(2): 140.
- Chen, Y., Z. Yin, and Q. Xie. 2014. Suggestions to ameliorate the inequity in urban/rural allocation of healthcare resources in China. *International Journal for Equity in Health* 13(1): 1–6.
- Copeland, V. C. 2005. African Americans: Disparities in health care access and utilization. *Health & Social Work* 30(3): 265–270.
- Dai, D. 2010. Black residential segregation, disparities in spatial access to health care facilities, and late-stage breast cancer diagnosis in metropolitan Detroit. *Health & Place* 16(5): 1038–1052.
- Delamater, P. L. 2013. Spatial accessibility in suboptimally configured health care systems: A modified two-step floating catchment area (M2SFCA) metric. *Health & Place* 24: 30–43.
- Ding, J., X. Hu, X. Zhang, L. Shang, M. Yu, and H. Chen. 2018. Equity and efficiency of medical service systems at the provincial level of China's mainland: A comparative study from 2009 to 2014. *BMC Public Health* 18(1): 214.
- Evans, R., S. Larkins, T. Cheffins, R. Fleming, K. Johnston, and M. Tennant. 2017. Mapping access to health services as a strategy for planning: Access to primary care for older people in regional Queensland. *Australian Journal of Primary Health* 23(2): 114–122.
- Florida, R., and C. Mellander. 2015. *Segregated city: The geography of economic segregation in America's metros*. Toronto, ON: Martin Prosperity Institute.
- Google Developers. 2021. *The Directions API overview*. <https://developers.google.com/maps/documentation/directions/overview>.
- Graves, B. A. 2008. Integrative literature review: A review of literature related to geographical information systems, healthcare access, and health outcomes. *Perspectives in Health Information Management* 5: 11.
- Guagliardo, M. F., C. R. Ronzio, I. Cheung, E. Chacko, and J. G. Joseph. 2004. Physician accessibility: An urban case study of pediatric providers. *Health & Place* 10(3): 273–283.
- Hansen, W. G. 1959. How accessibility shapes land use. *Journal of the American Institute of Planners* 25(2): 73–76.
- Hewko, J., K. E. Smoyer-Tomic, and M. J. Hodgson. 2002. Measuring neighbourhood spatial accessibility to urban amenities: Does aggregation error matter? *Environment and Planning A* 34(7): 1185–1206.
- Higgs, G. 2004. A literature review of the use of GIS-based measures of access to health care services. *Health Services and Outcomes Research Methodology* 5(2): 119–139.
- Huang, M., D. Luo, Z. Wang, Y. Cao, H. Wang, F. Bi, Y. Huang, and L. Yi. 2020. Equity and efficiency of maternal and child health resources allocation in Hunan Province, China. *BMC Health Services Research* 20: 1–10.

- Jamtsho, S., and R. J. Corner. 2014. Evaluation of spatial accessibility to primary healthcare using GIS. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences* 2(2): 79.
- Jankov, P., and C. Caref. 2017. Segregation and inequality in Chicago Public Schools, transformed and intensified under corporate education reform. *Education Policy Analysis Archives* 25: 56.
- Jin, C., J. Cheng, Y. Lu, Z. Huang, and F. Cao. 2015. Spatial inequity in access to healthcare facilities at a county level in a developing country: A case study of Deqing County, Zhejiang, China. *International Journal for Equity in Health* 14(1): 1–21.
- Kain, J. F. 1968. Housing segregation, negro employment, and metropolitan decentralization. *The Quarterly Journal of Economics* 82(2): 175–197.
- Kwan, M.-P. 1998. Space-time and integral measures of individual accessibility: A comparative analysis using a point-based framework. *Geographical Analysis* 30(3): 191–216.
- Langford, M., G. Higgs, and R. Fry. 2016. Multi-modal two-step floating catchment area analysis of primary health care accessibility. *Health & Place* 38: 70–81.
- Liu, D., and M.-P. Kwan. 2020. Measuring spatial mismatch and job access inequity based on transit-based job accessibility for poor job seekers. *Travel Behaviour and Society* 19: 184–193.
- Liu, D., M.-P. Kwan, and Z. Kan. 2021a. Analysis of urban green space accessibility and distribution inequity in the City of Chicago. *Urban Forestry & Urban Greening* 59: 127029.
- Liu, D., M.-P. Kwan, and Z. Kan. 2021b. Assessing job-access inequity for transit-based workers across space and race with the Palma ratio. *Urban Research & Practice* <https://doi.org/10.1080/17535069.2021.1923795>.
- Liu, J., S. Nambisan, X. Li, and X. Fu. 2020. Are young Americans carless across the United States? A spatial analysis. *Transportation Research Part D: Transport and Environment* 78: 102197.
- Loo, B. P., and W. W. Y. Lam. 2012. Geographic accessibility around health care facilities for elderly residents in Hong Kong: A microscale walkability assessment. *Environment and Planning B: Planning and Design* 39(4): 629–646.
- Luo, J. 2014. Integrating the Huff model and floating catchment area methods to analyze spatial access to healthcare services. *Transactions in GIS* 18(3): 436–448.
- Luo, W. 2004. Using a GIS-based floating catchment method to assess areas with shortage of physicians. *Health & Place* 10(1): 1–11.
- Luo, W., and F. Wang. 2003. Measures of spatial accessibility to health care in a GIS environment: Synthesis and a case study in the Chicago region. *Environment and Planning B: Planning and Design* 30(6): 865–884.
- Luo, W., and T. Whippo. 2012. Variable catchment sizes for the two-step floating catchment area (2SFCA) method. *Health & Place* 18(4): 789–795.
- Luo, W., and Y. Qi. 2009. An enhanced two-step floating catchment area (E2SFCA) method for measuring spatial accessibility to primary care physicians. *Health & Place* 15(4): 1100–1107.
- Mao, L., and D. Nekorchuk. 2013. Measuring spatial accessibility to healthcare for populations with multiple transportation modes. *Health & Place* 24: 115–122.
- Mayaud, J. R., M. Tran, and R. Nuttall. 2019. An urban data framework for assessing equity in cities: Comparing accessibility to healthcare facilities in Cascadia. *Computers, Environment and Urban Systems* 78: 101401.
- Moore, N. Y. 2016. *The south side: A portrait of Chicago and American segregation*. New York, NY: St. Martin's Press.
- Oppong, J. R., and M. J. Hodgson. 1994. Spatial accessibility to health care facilities in Suhum District, Ghana. *The Professional Geographer* 46(2): 199–209.
- Paez, A., C. D. Higgins, and S. F. Vivona. 2019. Demand and level of service inflation in Floating Catchment Area (FCA) methods. *PloS One* 14(6): e0218773.
- Paez, A., R. G. Mercado, S. Farber, C. Morency, and M. Roorda. 2010. Accessibility to health care facilities in Montreal Island: An application of relative accessibility indicators from the perspective of senior and non-senior residents. *International Journal of Health Geographics* 9(1): 1–15.
- Páez, A., R. G. Mercado, S. Farber, C. Morency, and M. Roorda. 2010. Relative accessibility deprivation indicators for urban settings: Definitions and application to food deserts in Montreal. *Urban Studies* 47(7): 1415–1438.
- Páez, A., D. M. Scott, and C. Morency. 2012. Measuring accessibility: Positive and normative implementations of various accessibility indicators. *Journal of Transport Geography* 25: 141–153.
- Pan, X., M.-P. Kwan, L. Yang, S. Zhou, Z. Zuo, and B. Wan. 2018. Evaluating the accessibility of healthcare facilities using an integrated catchment area approach. *International Journal of Environmental Research and Public Health* 15(9): 2051.
- Richardson, L. D., and M. Norris. 2010. Access to health and health care: How race and ethnicity matter. *Mount Sinai Journal of Medicine: A Journal of Translational and Personalized Medicine* 77(2): 166–177.
- Rosero-Bixby, L. 2004. Spatial access to health care in Costa Rica and its equity: A GIS-based study. *Social Science & Medicine* 58(7): 1271–1284.
- Sanchez, T. W., R. Stolz, and J. S. Ma. 2004. Inequitable effects of transportation policies on minorities. *Transportation Research Record* 1885(1): 104–110.
- Schuurman, N., M. Berube, and V. A. Crooks. 2010. Measuring potential spatial access to primary health care physicians using a modified gravity model. *The Canadian Geographer* 54(1): 29–45.
- Shah, T. I., S. Bell, and K. Wilson. 2016. Spatial accessibility to health care services: Identifying under-served neighbourhoods in Canadian urban areas. *PloS One* 11(12): e0168208.
- Shen, Q. 1998. Location characteristics of inner-city neighborhoods and employment accessibility of low-wage workers. *Environment and Planning B: Planning and Design* 25(3): 345–365.
- Sibley, L. M., and J. P. Weiner. 2011. An evaluation of access to health care services along the rural-urban continuum in Canada. *BMC Health Services Research* 11(1): 20.
- Smith, J. L., Z. Sonmez, and N. Zettel. 2021. Growing income inequality and socioeconomic segregation in the Chicago region. *Urban Socio-Economic Segregation and Income Inequality* https://doi.org/10.1007/978-3-030-64569-4_18.
- Tomer, A. 2011. *Transit access and zero-vehicle households*. Washington, DC: Brookings Institution.
- United States Census Bureau. 2018. *American Community Survey 5-Year Estimates*. <https://data.census.gov/cedsci/advanced?y=2018>.
- Wan, N., B. Zou, and T. Sternberg. 2012. A three-step floating catchment area method for analyzing spatial access to health services. *International Journal of Geographical Information Science* 26(6): 1073–1089.

- Widener, M. J., S. Farber, T. Neutens, and M. Horner. 2015. Spatiotemporal accessibility to supermarkets using public transit: An interaction potential approach in Cincinnati, Ohio. *Journal of Transport Geography* 42: 72–83.
- WHO (World Health Organization). 2019. *World health statistics 2019: Monitoring health for the SDGs, sustainable development goals*. Geneva, Switzerland: WHO.
- Wong, H. T., Y. Q. Guo, M. Y. L. Chiu, S. Chen, and Y. Zhao. 2016. Spatial illustration of health-care workforce accessibility index in China: How far has our 2009 health-care reform brought us? *Australian Journal of Rural Health* 24(1): 54–60.
- Xiong, X., C. Jin, H. Chen, and L. Luo. 2016. Using the fusion proximal area method and gravity method to identify areas with physician shortages. *PloS One* 11(10): e0163504.
- Zhang, X., M. E. Dupre, L. Qiu, W. Zhou, Y. Zhao, and D. Gu. 2017. Urban-rural differences in the association between access to healthcare and health outcomes among older adults in China. *BMC Geriatrics* 17(1): 151.
- Zhang, Y., Q. Wang, T. Jiang, and J. Wang. 2018. Equity and efficiency of primary health care resource allocation in mainland China. *International Journal for Equity in Health* 17(1): 140.